

Assignment Code: DA-AG-009

EDA | Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Dataset Link (Bike Details Dataset):

https://drive.google.com/file/d/1iKy23bMtEQShF_weneRNnYrFmzvpPOI3/view?usp=drive_link

- Download the Bike Details dataset.
- Complete all practical questions using Python and relevant data science libraries.
- Present your findings with appropriate visualizations and summary statistics.
- Save as PDF and submit.

Total Marks: 200

Question 1: Read the Bike Details dataset into a Pandas DataFrame and display its first 10 rows.

(Show the shape and column names as well.)

(Include your Python code and output in the code box below.)

Answer:

Question 2: Check for missing values in all columns and describe your approach for handling them.

(Include your Python code and output in the code box below.)

Answer:

Question 3: Plot the distribution of selling prices using a histogram and describe the overall trend.

(Include your Python code and output in the code box below.)

Answer:

Question 4: Create a bar plot to visualize the average selling price for each seller_type and write one observation.

(Include your Python code and output in the code box below.)

Answer:

Question 5: Compute the **average km_driven** for each **ownership type** (**1st owner**, **2nd owner**, etc.), and present the result as a bar plot.

(Include your Python code and output in the code box below.)

Answer:

Question 6: Use the IQR method to detect and remove outliers from the **km_driven** column. Show before-and-after summary statistics.

(Include your Python code and output in the code box below.)

Answer:

Question 7: Create a scatter plot of **year** vs. **selling_price** to explore the relationship between a bike's age and its price.

(Include your Python code and output in the code box below.)

Answer:

Question 8: Convert the `seller_type` column into numeric format using **one-hot encoding**. Display the first 5 rows of the resulting DataFrame.

(Include your Python code and output in the code box below.)

Answer:

Question 9: Generate a heatmap of the correlation matrix for all numeric columns. What correlations stand out the most?

(Include your Python code and output in the code box below.)

Answer:

Question 10: Summarize your findings in a brief report:

- What are the most important factors affecting a bike's selling price?
- Mention any data cleaning or feature engineering you performed.

(Include your Python code and output in the code box below.)

Answer: